

Important Drivers of AI on Edge

- 1. Affordable cost.** The product and the technology has to be suitable at a reasonable price-point, both for general consumers and enterprises
- 2. Strong use-cases.** Relates to devices having intelligence rather than just control features. Examples include smart wearables and Alexa
- 3. Mobility.** Ease of carrying around a device such as Alexa Echo Dot, which can be kept in a pocket
- 4. Speed.** Reduced latency and fast response time
- 5. Security of data.** It is the biggest concern. So people prefer storing data on edge rather than on cloud
- 6. Reduced downtime.** More downtime = More financial losses. This can be avoided with the help of preventive maintenance
- 7. Robustness and reliability of a machine**
- 8. Cost of implementing AI** on an edge device (infra-investment) and how effective it is going to be
- 9. Conditions needed to set up and run AI** on edge applications (Does it have to be battery-powered or not?)

Note. Many other drivers/parameters control the adoption of AI on edge. But it generally differs from one use case, or one industry, to another.

twenty years back. It was then that cloud brought inexpensive computing and started giving way to AI and machine learning (ML) workloads to be run on the cloud.

As we all know, IoT technology refers to devices having the ability to talk to the Internet, push data to the cloud, and also have the ability to control or get controlled remotely. AI is a set of frameworks that makes these devices capable of taking their own decisions. Edge computing refers to when the devices can take self-decisions or have local servers through which data can be processed.

"As computing became cheaper, particularly with supercomputers in the form of smartphones and various IoT devices at our expense, the need to bring in that type of intelligence onto the edge became relevant," as per Venkateswar Melachervu, CTO, Brillium Technologies. He says, "Edge is where data resides and is collected."

AI on edge is a system that has AI capabilities on the device itself or at some local node. This innovative feature allows devices to make their own decisions based on their previous learnings of different situations and their outcomes.

For instance, devices such as smartphones, smartwatches—even a temperature sensor—can be classified as edge devices. It is to be understood that edge computing is needed not

just for the industry but also for the consumer.

"A lot of data is driven from machines or IoT devices, and that data needs to be computed from the security and privacy point of view. Therefore we need to have a better understanding of the differences between IoT, cloud, and edge computing," opines Dharmendra Kumar, head of IoT, Arihant.

Since intelligence on edge devices such as smartphones, smartwatches, and connected cars has been there for a long time, Rishi Gargi, founder, Thingify adds another important point, "A boundary needs to be established between what is AI on edge and what other technologies are, and how AI on edge is leveraging AI without having any Internet or making decisions on its own."

Why go for AI on edge

The power of today's computing is much more than what was available earlier. Even the latency and longevity of the batteries have improved. All these essentially have helped AI to be pushed for running on the edge, be it IoT devices, within the gateways, or the edge network.

Edge computing generally incorporates a processor and storage, as well as logic to process information. An example is Alexa, which uses natural language processing (NLP) at the edge to respond to queries from a human

being. But AI on edge is more than just computing. "It is not just about processing the data, but also deriving trends and patterns or intelligence by applying machine learning algorithms to give immediate feedback to the user," thinks Vinay Solanki, head of IoT, Napino.

An example could be of a smart camera that has been installed in a smart city application for monitoring vandalism. While capturing videos or images, it could decide locally without sending any feedback to the cloud and thus become smarter on its own.

Data security is the biggest concern

The real challenge for a customer is regarding data security. People do not want to save their data on the cloud. When running an algorithm in the cloud on a stored data, it should be properly safeguarded, both on the device from where it has been generated and in the network. Cloud can enforce certain other security protocols like firewall, some encryption-decryption logic, or public-private key for the handshake, and make it safer.

"Some users might not be willing to share their data on the cloud. So, this approach can transform the capabilities of such applications to store data on mobile as well as process it. Health-related applications are extremely sensitive, therefore it is important to process the data on the mobile," believes Amit Kurhekar, head of Data Science, Yodlee.

"It may sound a little oxymoronic because when you have distributed computing resources in the edge network, you are exposed to data attack (for instance DDoS). But usually, AI-edge is considered more secure for your data," as per Venkateswar Melachervu.

Nikhil Bhaskaran believes that since people do not want all of their private data to go on the cloud, the edge is therefore becoming popular.

The cost concern

Having AI on cloud vs AI on edge is an important question, because on the edge it is the responsibility of the device and the device designer to ensure that it is secured. If some-